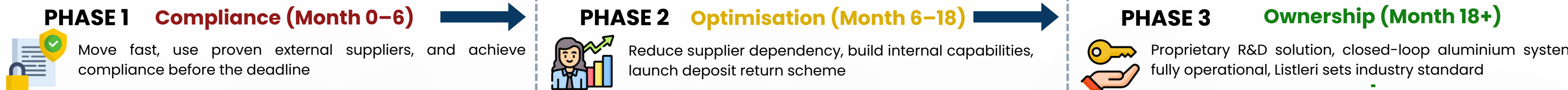


# Listleri Packaging Transition Strategy

Eliminating single-use plastic across all SKUs without eroding market leadership



**STILL WATER**

**Solution: RyPax Fibre Bottle**  
Scales in 6 months (partial)

|                  |              |
|------------------|--------------|
| Supplier:        | RyPax (Asia) |
| Shelf life:      | 6–12 months  |
| Cost vs plastic: | +20–30%      |
| Recovery:        | 9–12 months  |

**Why it works:** Curran nanocellulose + bamboo + bagasse: OTR <1.0 ml/m²/day (still water), no CO₂ needed, fully biodegradable, no plastic liner, low-cost inputs.

**Bridge suppliers:** ITC-Frugalpac partnership covers volume gap during RyPax ramp-up.

**CARBONATED DRINKS**

**Solution: Aluminium Cans + Closed Loop**  
Scales immediately via Ball Corp India

|                    |                            |
|--------------------|----------------------------|
| Supplier:          | Ball Corp India            |
| Scrap net per can: | ₹0.50–1                    |
| Cost vs plastic:   | +100–150% → self-financing |
| Recovery:          | 18–24 months               |

**Why aluminium wins:** Scalable for CO₂, infinitely recyclable, 95% less energy to recycle, higher pallet efficiency than glass, more premium than plastic.

**Why others fail:** Paper bottles can't hold CO₂. PLA bioplastics lose carbonation. Glass is heavy, fragile, and costly to transport.

**CLOSED LOOP**  
sale market → returns retailer → Retailer sells scrap new cans (60 day) ← Listleri collects → Ball/Hindalco

**JUICE**

**Solution: Tetrapak Cartons**  
Scales fast. Infra already exists in India.

|                  |                   |
|------------------|-------------------|
| Supplier:        | Tetrapak India    |
| Shelf life:      | 12+months ambient |
| Cost vs plastic: | +10% at capacity  |
| Recovery:        | 6–9 months        |

**Why Tetrapak:** Lowest cost delta across SKUs. Fully operational in India-zero setup delay. Aseptic packaging delivers 12+ month shelf life without refrigeration. Proven at scale (Frooti, Paper Boat) with strong consumer acceptance.

**Why not RyPax for juice:** Aseptic barrier required for juice shelf stability, RyPax validated for still water only

**R&D: PHASE 3 MOAT**

**Nanocellulose (CNC/CNF):** strength + natural decay. PLA: moisture barrier + water resistance. PBAT: flexibility, prevents cracking. **Hydrophobic coating:** reduces water vapour transmission ~200x

**Pure cellulose** degrades in 28–60 days. PLA alone is brittle. **The blend solves both. Bio Elite India** already manufactures on this principle; domestic proof of concept exists.

Fund from Phase 1 savings → develop formulation  
manufacture in-house ← file patents (Year 2)  
(Phase 3) license to other FMCGs → new revenue stream.

Competitors scramble for suppliers. Listeri sells them the solution

**Secondary Packaging: Bambrew**

Bengaluru-based startup, Compostable packaging from bamboo and plant. CPCB certified, biodegrades in 130 days.

- Replaces plastic shrink wrap, and shipping cartons
- CPCB certified (130 days)
- Pure OpEx | Zero CapEx
- Solves the secondary packaging layer end-to-end

**FULL COST OF TRANSITION**

|                                |                       |
|--------------------------------|-----------------------|
| Juice (Tetrapak):              | +10% per unit         |
| Carbonated (Aluminium):        | +100–150% per unit    |
| Still water (RyPax):           | +20–30% per unit      |
| RyPax line modification:       | Medium one-time capex |
| Secondary (Bambrew):           | Minimal - OpEx only   |
| Reverse logistics setup:       | Medium one-time capex |
| Tetrapak Phase 1 (outsourced): | Zero capex            |

- LOSSES DURING SWITCHOVER**
- Inventory write-off:** Non-compliant PET stock written off, with the heaviest in carbonated.
  - Production downtime:** 2–4 weeks, reduced output during line modifications (months 1–3).
  - Advance payments:** Capacity locking with Ball Corp + RyPax = upfront cash drain.
  - Margin compression:** Cost delta absorbed months 1–6 to protect market share.
  - Kirana friction:** 3–6 months shelf adjustment, volume dip, no structural loss.
  - RyPax ramp gap:** Bridge suppliers cost more per unit, temporary margin drag.

